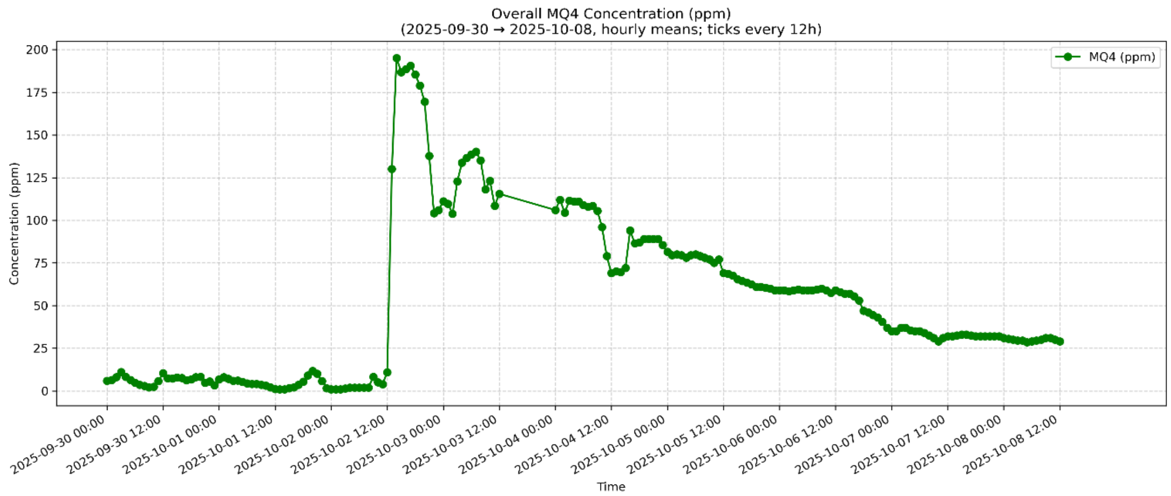
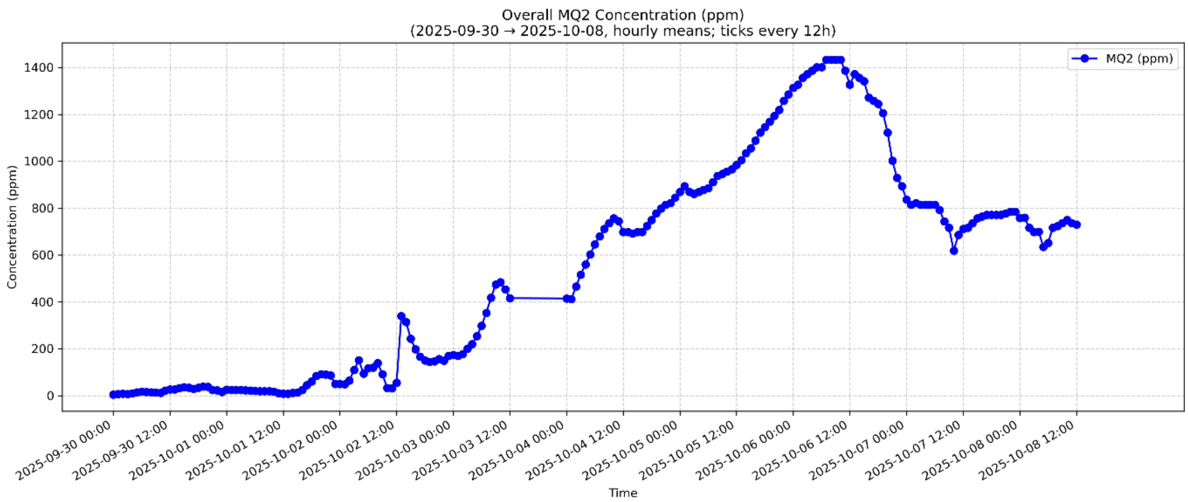
Experiment 1graph:

- **Y-axis (PPM):** Parts per million; higher ppm = more target gas.
- **Pipeline:** ADC → Vout → Rs = RL·(5−Vout)/Vout → (Rs/R0) → PPM.
- **Terms: Rs** = sensor resistance (drops with more gas); **R0** = clean-air baseline (from calibration); **RL** = fixed load resistor.
- **Meaning: Rs/R0** shows overall gas effect (mixture). **PPM** is a per-gas estimate using a fitted curve.
- **Curves used:**

  MQ-2 ≈ 50·(Rs/R0)^−2.3 (hydrocarbons/Ethylene);

  MQ-4 ≈ 1000·(Rs/R0)^−2.95 (Methane).
- **Reading the graph:** More gas → lower Rs/R0 → higher ppm. **Peaks** = gas bursts; **flat/low** = clean or stable air.

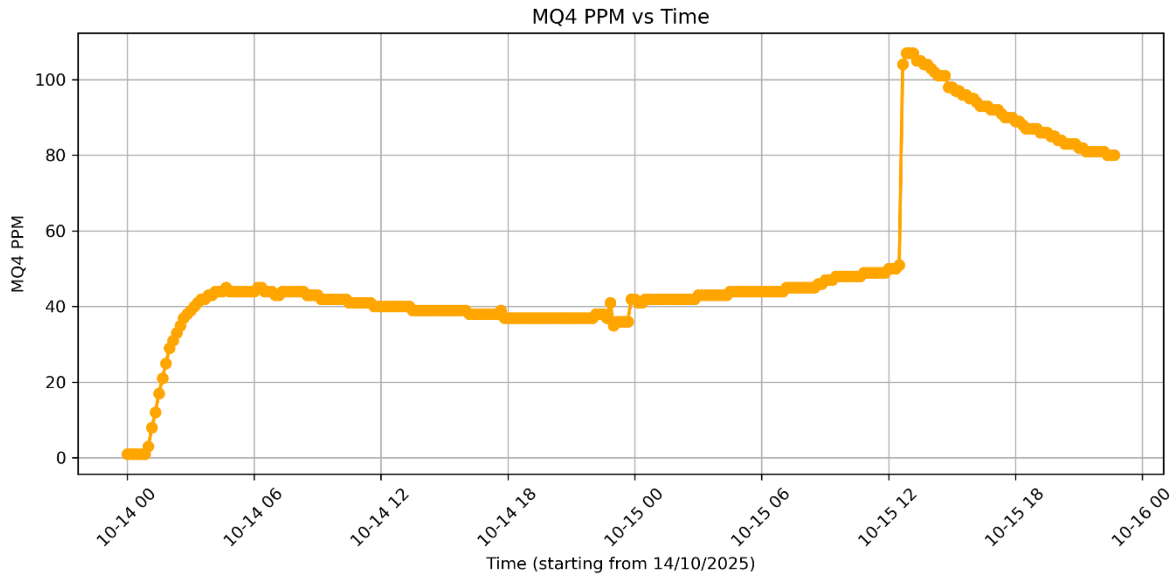


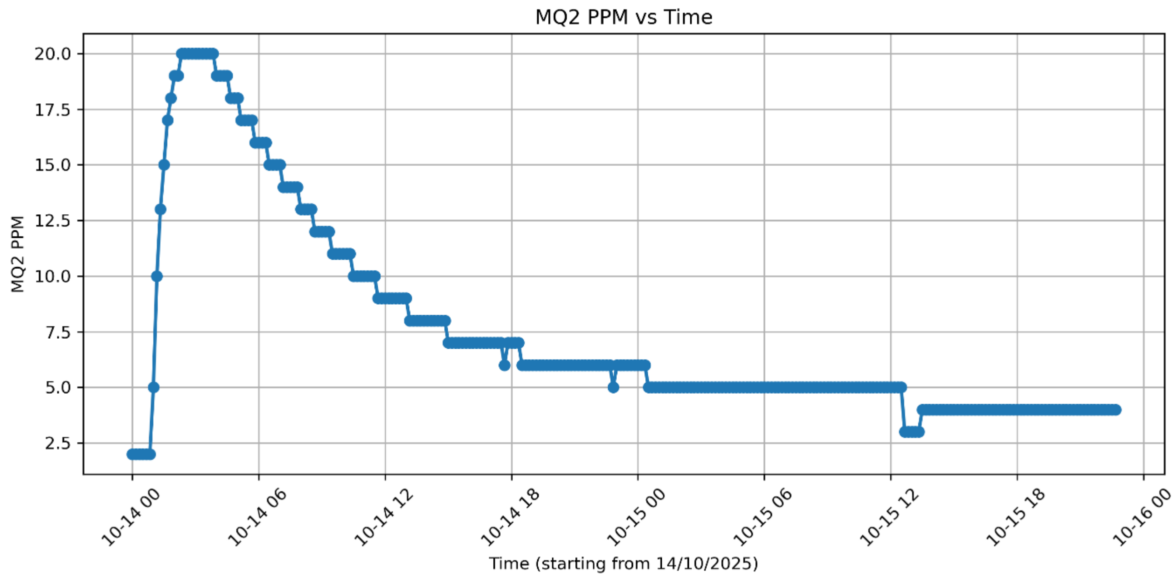


Experiment 2 graph:

- **Y-axis (PPM):** Parts per million; higher ppm = more target gas.
- **Pipeline:** ADC → Vout → Rs = RL·(5−Vout)/Vout → (Rs/R0) → PPM.
- **Terms: Rs** = sensor resistance (drops with more gas); **R0** = clean-air baseline (from calibration); **RL** = fixed load resistor.
- **Meaning: Rs/R0** shows overall gas effect (mixture). **PPM** is a per-gas estimate using a fitted curve.
- **Curves used:**

  MQ-2 ≈ 50·(Rs/R0)^−2.3 (hydrocarbons/Ethylene); MQ-4 ≈ 1000·(Rs/R0)^−2.95 (Methane).
- **Reading the graph:** More gas → lower Rs/R0 → higher ppm. **Peaks** = gas bursts; **flat/low** = clean or stable air.





Experiment 3:

• **Y-axis** (Gas Index 0–100): Higher index = more target gas.
• **Pipeline**: ADC → Vout → Rs = RL·(5/Vout − 1) → (Rs/R0) → GasIndex.
• **Terms:** Rs = sensor resistance (drops with more gas); R0 = clean-air baseline (from calibration); RL = fixed load resistor.
• **Meaning:** Rs/R0 captures overall gas effect (mixture). GasIndex is a normalized, human-friendly scale (not gas-specific ppm).
• **Mapping used:**
RS/R0 ≥ 1.0 → GI=0; 0.6→≈40; 0.3→≈80; ≤0.1→100 .
• Reading the graph: More gas → lower Rs/R0 → higher GasIndex.